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Geospatial Query System with Natural Language Processing and GIS Integration

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Abstract

Choosing where to open a business usually means weeks of manual surveys and scattered research across maps, listings and reports. This project presents a natural-language spatial intelligence platform for real estate and business site selection in Mumbai. A user asks a question in plain English, such as "safest area near a metro for a cafe", and an LLM pipeline translates it into verified spatial SQL over a live PostGIS database of roads, points of interest, real estate listings, crime zones and transit stops. The answer is returned as an interactive map with a short explanation, the generated SQL and a data-grounded budget estimate, turning a multi-week survey into a single instant query.

Background

Traditional GIS tools demand expertise in spatial data, SQL and specialised software, which keeps location intelligence out of reach for most businesses. Recent work on LLM-based GIS and spatial Text-to-SQL (LLM-Geo, NALSpatial, GS-SQL, GeoLLM) is promising, but still lacks query testing and logging, depends on predefined mapping rules, and struggles with multi-entity spatial reasoning and distance errors. Our system addresses these gaps with schema-grounded intent parsing, multi-step query planning, metre-accurate distance handling and a validation loop before execution, all grounded in real city data.

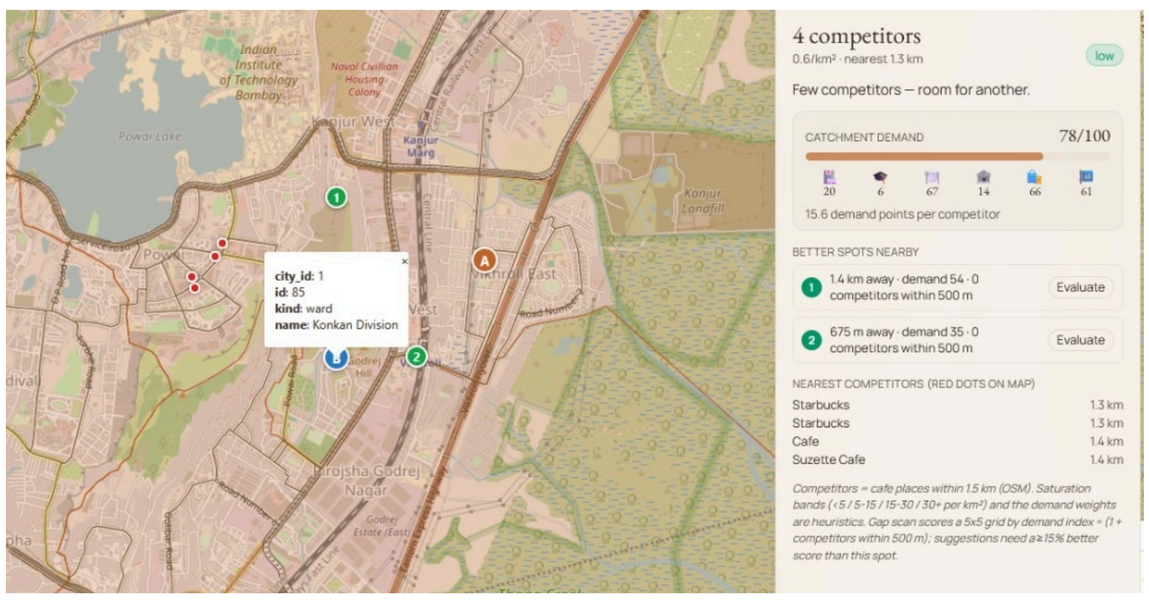


Figure 1: GeoQuery workspace: competition and catchment analysis for a cafe at Vikhroli

Objectives

- Simplify site selection so businesses can find and evaluate new locations without manual survey work.
- Generate automatic, data-grounded cost estimates (rent, setup, licenses) for any chosen spot.
- Compare locations side by side on nearby comparables, footfall, amenity density, safety and transit access.
- Replace scattered research across sources with a single natural-language query.
- Save time and cost with instant, grounded answers instead of weeks of on-ground survey.

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Methodology

A FastAPI backend receives the plain-English query and passes it to an LLM pipeline that classifies intent, grounds it in the database schema, plans the query and generates spatial SQL. The SQL runs on a PostGIS database, and a validation and correction loop re-plans any query that fails or returns empty results. Validated results are converted to GeoJSON and rendered on an interactive MapLibre map.

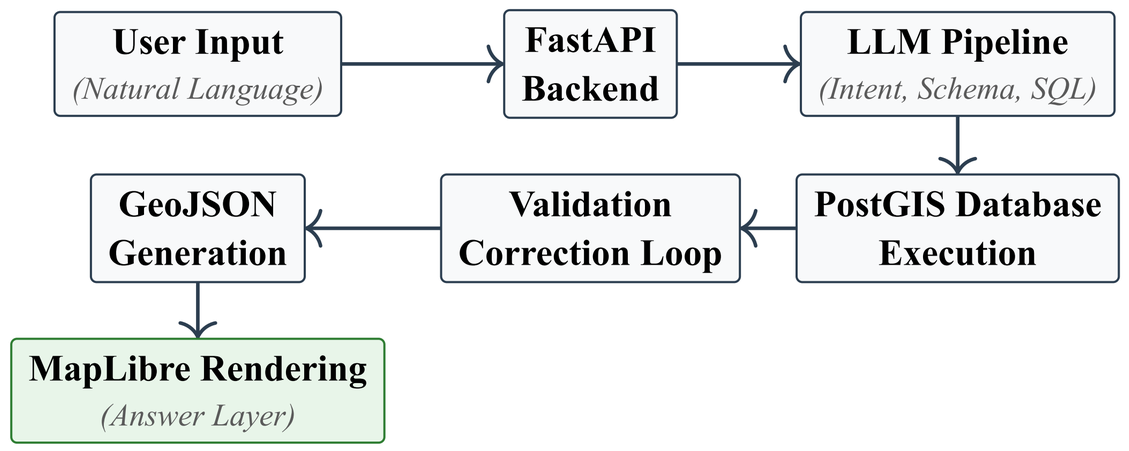


Figure 2: Proposed Methodology

Results and Discussion

The platform runs on live Mumbai data: 48 administrative boundaries, 16,800+ road segments, 10,800+ points of interest, 55,600+ real estate listings, 2,200+ transit stops and 40 crime-safety zones. Answers arrive in seconds with a map, an explanation and the SQL behind them. In evaluation, it estimated about ₹62 lakh to open a café in Bandra and about ₹1.3 crore for a premium restaurant in Lower Parel.

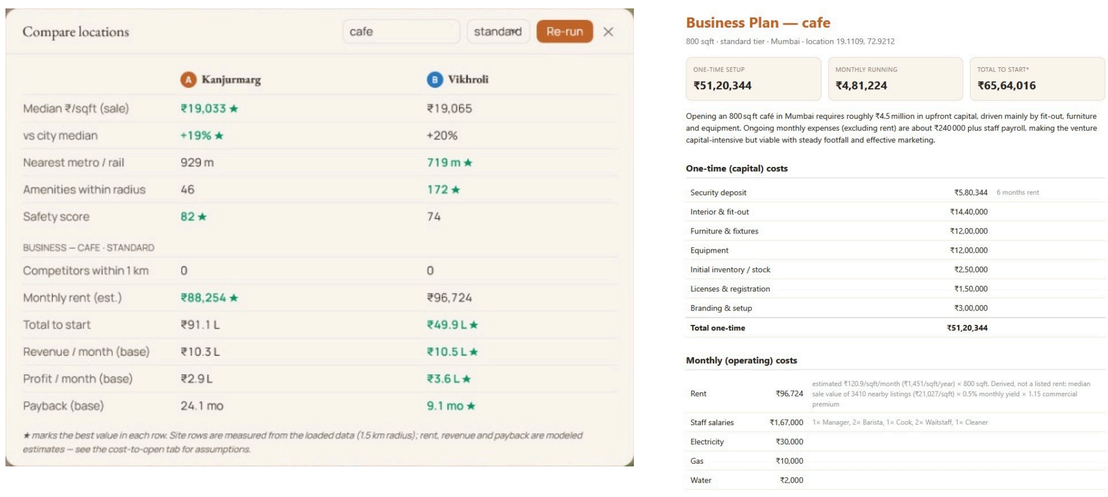


Figure 3: Location comparison (Kanjurmarg vs Vikhroli) and cost-to-open report for a cafe

Conclusion

Natural-language spatial intelligence makes location decisions accessible to businesses without GIS or data teams. One verifiable query interface replaces weeks of scattered research, and the city-agnostic design allows extension well beyond Mumbai.

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